

Probability Distributions

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What is a probability distribution?

- Mathematical model that characterizes a *population*
- Describes all possible values of a random variable
- Describes likelihood associated with each value

Main categories of probability distribution

1. Discrete probability distributions

- Binomial distribution
- Poisson distribution

2. Continuous probability distributions

- Normal distribution
- t distribution
- F distribution
- Chi square distribution

Two rules of a valid probability distribution

- Probability of any event cannot be negative
- Total sum of all probabilities = 1 (Total area under the curve = 1)

Binomial distribution

- Describes the number of successes in a fixed number of trial (n)
- Each trial has only two *mutually exclusive* outcomes (p,1-p)
- Used in analysing proportions or risk (dead/alive; treatment success/fail)
- *Probability* of success (p) must be *constant* for every trial
- *Outcome* of one trial must be *independent* of the result of another trial

Parameters of binomial distribution

- Mean (measure of central tendency) is $\mu = n \times p$
- Variance (measure of spread) is $\sigma^2 = n \times p \times (1 - p)$
- Parameters are *number of trials* (n) and *probability of success* (p)

Binomial distribution formula

$$P(X = x) = {}_nC_x \cdot p^x \cdot (1 - p)^{n-x}$$

Example of binomial distribution

- Let's flip a fair coin for 10 times and the probability of head is p
- $p = 0.5; n = 10$
- $\mu = p \times n = 5$
- $\sigma^2 = n \cdot p \cdot (1 - p) = 2.5 ; \sigma = 1.581$

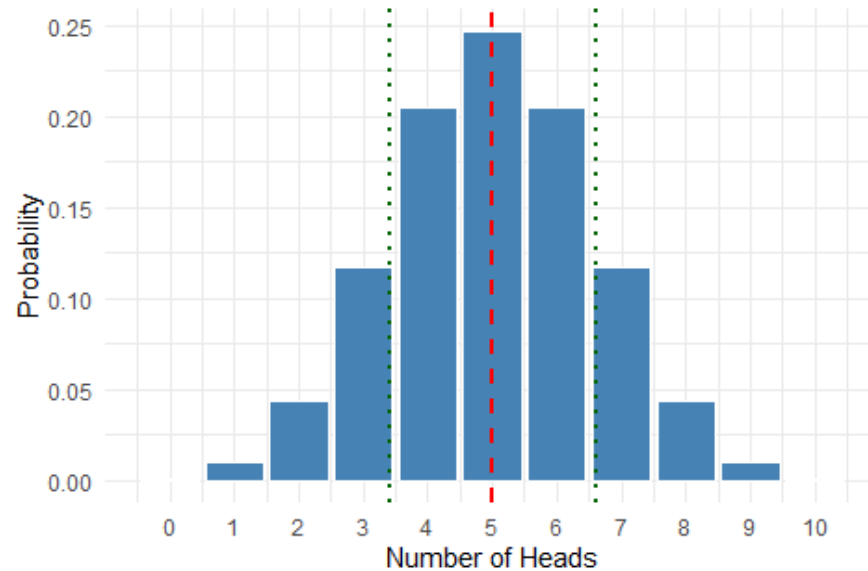
Data

x	y
0	0.0009766
1	0.0097656
2	0.0439453
3	0.1171875
4	0.2050781
5	0.2460938
6	0.2050781
7	0.1171875
8	0.0439453
9	0.0097656
10	0.0009766

Binomial Distribution

Binomial Distribution Curve of flipping a fair coin
($n = 10$, $p = 0.5$)

Mean (Red): 5 | Std Dev (Green): 1.6



Binomial distribution of an unfair coin

